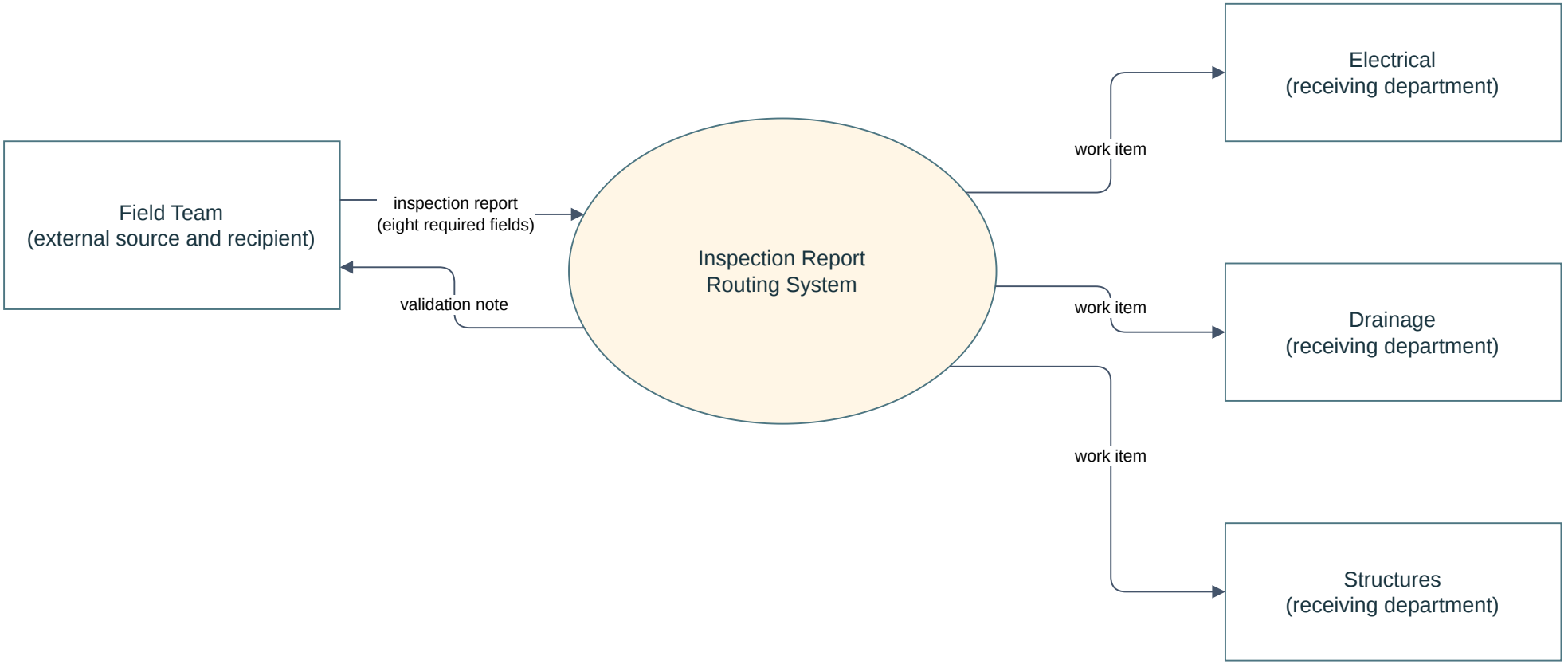
CIT 140 · Week 6 · System boundary

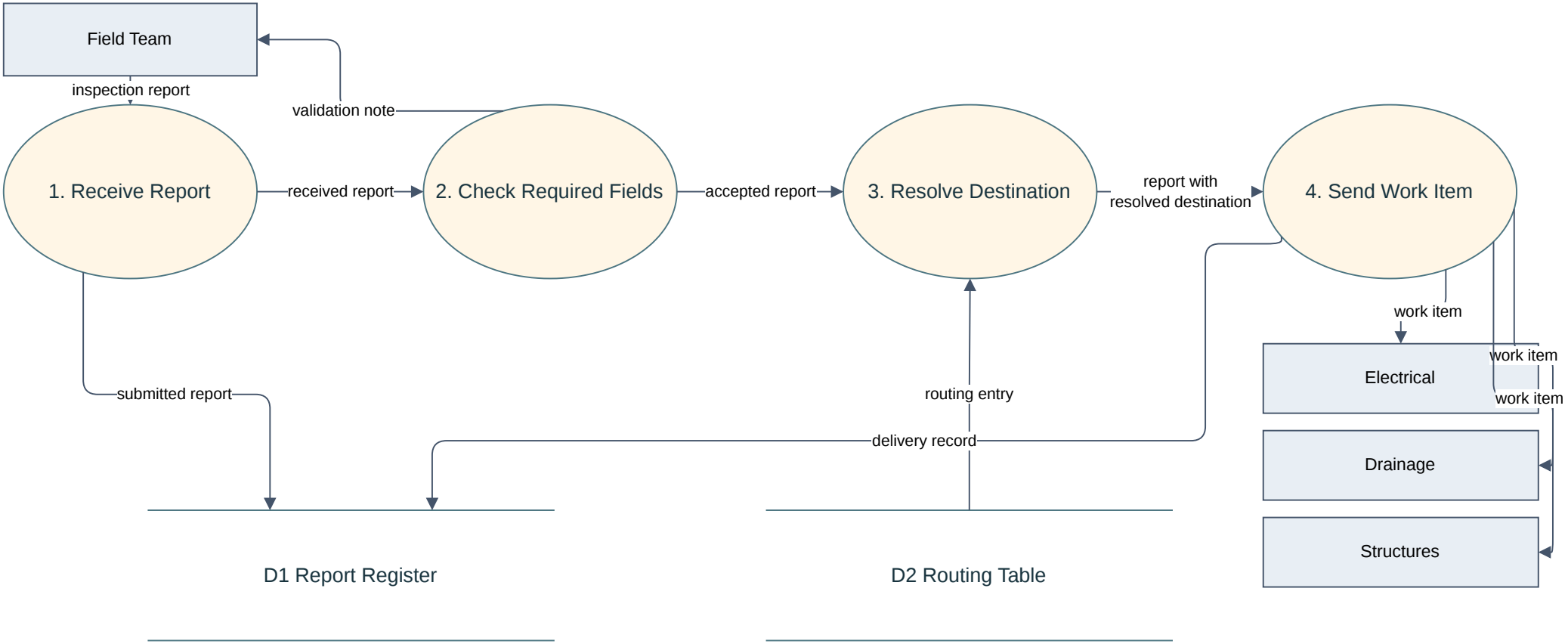
System modeled: the Inspection Report Routing System. The Field Team supplies inspection reports and receives validation notes. Electrical, Drainage and Structures receive routed work items. Every arrow names the data that crosses the boundary. Internal processes and stores appear on page 2. [W06-S01]



Inside the boundary: receiving the inspection report, checking the eight required fields, resolving the destination from the routing table, sending the work item, and the two internal stores D1 Report Register and D2 Routing Table.
Outside the boundary: the Field Team; Electrical, Drainage and Structures; drone flight control, physical inspections and repair work; and maintenance of the Routing Table, which is performed by a service administrator.
Sources: W06-S01 (boundary and processes), W06-S02 (report fields), W06-S03 (required destinations), W06-S04 (delivery log).

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Inside the Inspection Report Routing System: four processes and two data stores. Each arrow names the data that moves, not the order of steps. External inputs and outputs match page 1. [W06-S01]



Notes. P2 Check Required Fields tests the eight fields defined in W06-S02. A report that fails the check produces a validation note to the Field Team and does not continue to P3. P4 writes a delivery record to D1 Report Register; the delivery log in W06-S04 is an export of those records. The service administrator who maintains D2 Routing Table works outside this boundary, so table maintenance is not modelled here. [W06-S01]

CIT 140 · Week 6 · Definitions and finding

All eight required fields from W06-S02, with one traced routing discrepancy and the question the supplied records cannot answer.

Field	Meaning	Example	Allowed value / format	Source
report_id	Unique reference for one inspection report	CJ-601	CJ- followed by three digits	W06-S02
asset_id	Reference for the inspected asset	SL-101	Two-letter prefix plus three digits	W06-S02
asset_type	Kind of asset inspected; sets the required department	BRIDGE	STREETLIGHT, DRAIN or BRIDGE	W06-S02, W06-S03
finding_code	What the inspection found	SURFACE_DAMAGE	LAMP_OUT, BLOCKED or SURFACE_DAMAGE	W06-S02
priority	Urgency of the work; does not change the department	URGENT	ROUTINE or URGENT	W06-S02, W06-S03
captured_at	UTC time the inspection was captured	2026-10-14T09:10:00Z	YYYY-MM-DDTHH:MM:SSZ	W06-S02
photo_ref	Attachment reference for the image; no image files supplied	IMG-603	IMG- followed by three digits	W06-S02
survey_area	Fictional survey area of the city; not a coordinate	CENTRAL	NORTH, CENTRAL or SOUTH	W06-S02

Finding. Report **CJ-603** (asset_type BRIDGE, survey_area CENTRAL, ROUTINE) had an expected destination of **Structures** and an observed destination of **Electrical**. Fields from W06-S02; required destination from W06-S03; observed destination from the delivery log in W06-S04; process path from W06-S01. The same mismatch appears for CJ-604 and CJ-607, the other two BRIDGE reports, so no BRIDGE report in this export reached Structures, while every STREETLIGHT and DRAIN report reached its required department. Priority is not the factor: those three are one URGENT and two ROUTINE, and W06-S03 states that urgency does not change the department. All eight reports were ACCEPTED, so the eight required fields were present and the discrepancy arises after P2 Check Required Fields. This is an observed result, not an established cause.

Unanswered question. What routing entries did D2 Routing Table hold when these work items were sent? W06-S03 gives the required mapping but states it does not prove what the running table contained, and no runtime snapshot, deployment history or code was supplied. A dated export of D2 and its change history, which a service administrator maintains outside this boundary, would be needed to establish the cause. [W06-S01, W06-S03, W06-S04]